object oriented Programming using java - Home Assignment-1

T. Sai Akhila

1. Create a class student with the following data members:
   • student ID • student Name • Age • Email address • ph no.
2. Create methods to:
   • Accept student details
   • Display student details.

```
Import java.util.scanner;
 class student {
   int student ID;
   string student Name;
   int age;
    string email;
    string phone number;
  void accept Details() {
    Scanner sc = new scanner(System.in);
    System.out.println("Enter student ID: ");
    student Id = sc.nextInt();
     sc.next line();
     System.out.println("Enter Age:");
      age = sc.nextline();
        sc = nextline();
     System.out.println("Enter email address: ");
      email = sc.nextline();
      System.out.println("Enter phone number: ");
       phone number = sc.nextline();
     }
      void display Details() {
         system.out.println("\n.... Student Details... ");
          System.out.println("Student ID:"+ stu Id);
          system.out.println("Student Name:"+student Name);
         system.out.println("Age :"+ Age);
         System.out.println("Email Address:"+email);
         System.out.println("Phone number:" +ph.num);
      }
  }
   public class student information Management
     {
```

```
System.out.println("Enter marks of subject 5:");
// Calculate Total and Average
int total = m1+m2+m3+m4+m5;
double average = total/5.0;
System.out.println("\n Total marks =" + total);
System.out.println(" Average marks =" + avg);
//Determine Grade
    if (average >=90) {
    System.out.println(" Grade = A");
    }else if (average >=75){
    system.out.println("Grade = B");
    }else if (average >=60) {
      System.out.println("Grade = C");
    }else {
      System.out.println(" Grade = F");
    }
    //Display pass or Fail
      if (average >=50) {
      system.out.println("Result = Pass");
      }else {
        System.out.println("Result = Fail");
      }
    //Department using switch.
      System.out.println("\n Enter Department code (1-4):");
        int code = sc.nextInt();
        switch(code) {
      case-1:
        System.out.println("Department: CSE");
        break;
      case-2:
        System.out.println(" Department: CSE (AI&ML)");
        break;
      case-3:
        System.out.println("Department: CSE IT");
        break;
      case-4:
        System.out.println("Department: ECE");
        break;
```

```
public static void main(string[] args){
    Student.s = new Student();
     s.acceptDetails(sc);
     s.displayDetails();
    }
}
```

Sample output:

```
Enter student ID: 101
Enter student name: Akhila
 Enter Age: 22
 Enter ph.number: 9493852349
Enter email address: akhilachandra@gmail.com
  ---- Student Details ----
   Student ID: 101
  Student Name: Akhila
    Age : 22
  Email address: akhilachandra@gmail.com
   phone number: 9493852349.
```

- Part B: Operators and control statements.

3. Accept marks in five subjects and:
   - Calculate the total and average using arithmetic operators.
   - Determine the grade using if-else if statements.
   - Display pass if the avg is 50 (or) above, otherwise display file. Use a switch statement to display the students department based on the department code entered by the user • 1-CSE • 2-CSE(AI&ML) • 3-CSIT • 4-ECE
   - Default - INvalid department.

```
Import java.util.scanner;
 public static void main{
    scanner sc = new scanner(system.in);
    System.out.println("Enter marks of subject+1:);
       int m1 = sc.nextInt();
    system.out.println("Enter marks of subject+2:);
       int m2 = sc.nextInt();
    System.out.println("Enter marks of subject +3:);
       int m3 = sc.nextInt();
    System.out.println("Enter marks of subject +4:);
       int m4 = sc.nextInt();
```

```
default:
    System.out.println("Invalid department");
    }
    sc.close();
    }
}
```

Sample output:

```
Enter marks of sub1: 80
Enter marks of sub 2: 75
Enter marks of sub 3: 90
Enter marks of sub 4: 85
Enter marks of sub 5: 70
  Total marks = 400
  Avg marks = 80.0
    Grade = B
    Result = pass
Enter department code (1-4): 2
  Department: CSE (AIML)
```

- Part C: Arrays and strings

5. Store the marks of five subjects in an array and display
   - Highest mark
   - Lowest mark
   - Average mark.

6. Perform the following string operations on the student's Name
   - Convert to uppercase
   - Display the length of the name
   - check whether the name starts with a given character

```
import java.util.scanner
  public static void main(string[]args){
    scanner sc = new scanner(system.in);
    int highest = marks[0];
    int lowest = marks[0];
    for(int i=1; i<marks.length; i++){
      if(marks[i] > highest){
        highest = marks[i];
      }
      if(marks[i] < lowest){
```

```
            lowest = marks[i];
        }
    }
    System.out.println("\n---Array operations---");
    System.out.println("Highest Marks:" + highest);
    System.out.println("Lowest Marks:" + Lowest);
    System.out.println("Average Mark:" +Avg);
     sc.nextline();
     System.out.println("\n Enter character to check whether th
                name starts where it:");
     char given char = sc.nextline().charAt(0);
      System.out.println("\n--- string operations----");
      System.out.println("Name in uppercase:" + student
                        +Name to uppercase());
      System.out.println("Length of Name:" + student
                        student Name.length());
      System.out.println("First character:" + student
                        student+Name.charAt(0));
     if(student.student Name At(0) = given char) {
      System.out.println("Name starts with" " " + givenchar+
                        " ":Yes");
      }else {
     System.out.println("Name starts with " " + givenchar+"
                        :"No");
     } sc.close();
      }
    }
```

Sample output:

```
    Higherst mark:92
     Lowest mark:78
      Avg mark:84.6
 Enter a character to check whether the name starts with
 it's:s
---- string operation ----
Name in upper case:CHANDRA
 Length of Name:6
 First character:S
 Name starts with 's':Yes.
```

Part 0: theory

2500030391
T. Sai Akhila.

1. Procedural programming is Vs object oriented programming
Procedural programming and oops are two different approaches to delivery software in POP a program is organised mainly into functions or procedures that operate on data in oop, a program is organised around objects and classes that include (or) combine data and methods.
In oop data is generally seperated from functions which can make data less secure in oop, data access modify classes and methods are combine within classes, and access modifies such as private, public and protected provide better data security through encapsulation POP is suitable for smaller programs where the problem can be divided into functions. oop is particularly useful for large and complex applications because it organizes programs into manageable objects.
Therefore, POP focus mainly on functions and procedures, whereas oop focuses on objects, data and their intera -ctions.

2. Briefly explain four pillars of oops with examples.
The four main pillars are
1. Encapsulation: It means combining data and methods into a single data class and controlling access to the data. For example, a student class can keep student Id private and provide public class to class access.
2. Inheritance: It allows one class to acquire the properties and methods of another class. For example, A dog class can inherit common properties from an animal class.
3. Polymorphism: It means "many forms". The same method can behave differently depending on object. for example, sound() method can produce different results for dog on a cat.
4. Abstraction: It means hiding unecessary implementation det -ails & showing only essential features. for examples when using on ATM, the user performs operations such as with drawing money without knowing the internal processing.